Public Health Dashboard Report

Introduction

This public health dashboard is designed in such a way as to give users access to information on the trends in diseases and deaths and how the overall burden of health indicators globally. The use of this dashboard makes it easy for health officials, policymakers, and researchers to monitor diseases, assess the health indicators of various countries, and take decisions based on evidence.

The dashboard has used the interactive features of Power BI to present health data into visual representations through graphs, maps, Key Performance Indicators, and trend analyses..

Objectives

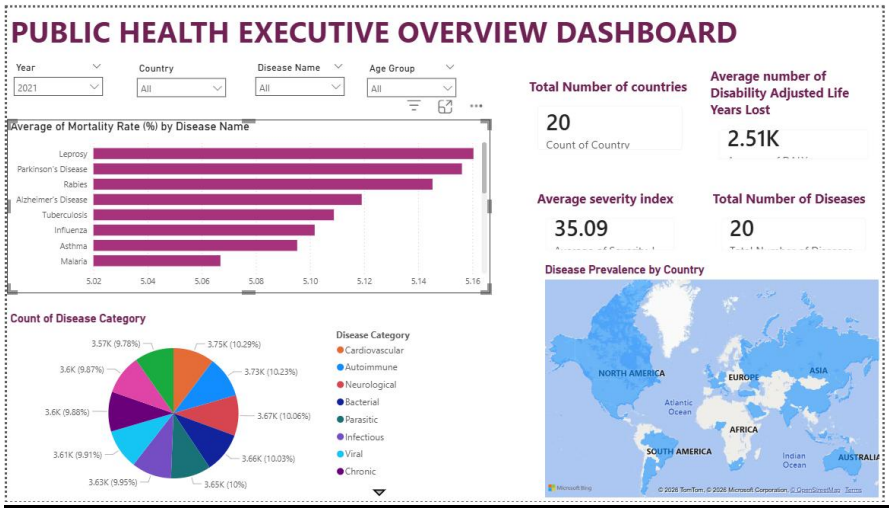
- To monitor disease prevalence and mortality trends across countries.
- To identify high-risk diseases and regions with severe health burdens.
- To support healthcare resource allocation and policy planning.
- To provide interactive exploration of public health data.
- To improve understanding of disease patterns over time.

Dataset description

The dataset contains public health information collected across multiple countries and years. It includes variables related to:

- Disease names
- Disease categories
- Mortality rates
- Incidence rates
- Severity index
- Disability Adjusted Life Years (DALY)
- Country information
- Age groups
- Temporal data (2000–2024)

Dashboard pages explanation



Page 1:

This page provides a high-level summary of the global public health situation between the year 2000 and 2024. It is designed to give decision-makers quick insights into disease burden and mortality patterns.

The various Key Performance indicators in the dashboard include:

1. Total Number of Countries: 20
2. Total Number of Diseases: 20
3. Average Severity Index: 35.09
4. Average DALY Lost: 2.51K

They allow users to quickly assess the overall health landscape without analysing detailed visuals.

(i) Average Mortality Rate (%) by Disease Name

A horizontal bar chart ranks diseases according to mortality impact.

Key observations include:

a) Leprosy recorded the highest mortality rate.
b) Parkinson's, Alzheimer's, and Rabies also contributed significantly to mortality.
c) Ebola and Dengue showed comparatively lower mortality values within this dataset.

(ii) Count of Disease Category

The pie chart displays the distribution of disease categories within the dataset.

It reveals a relatively balanced distribution across categories such as:

a. Cardiovascular
b. Autoimmune
c. Neurological

Each category contributes approximately 10% of the total dataset representation thus it provides an overview of the diversity and balance of disease classifications in the dataset.

(iii) Disease Prevalence by Country

The geographical map visualizes disease prevalence across different regions of the world.

The visualization includes regions like: Africa, Europe, Asia, North America, South America, Australia

Colour intensity is used to indicate prevalence levels, helping users identify geographical hotspots and regional disease burdens.

(iv) Interactive Features

The dashboard has slicers and filters that improve interactivity and user control.

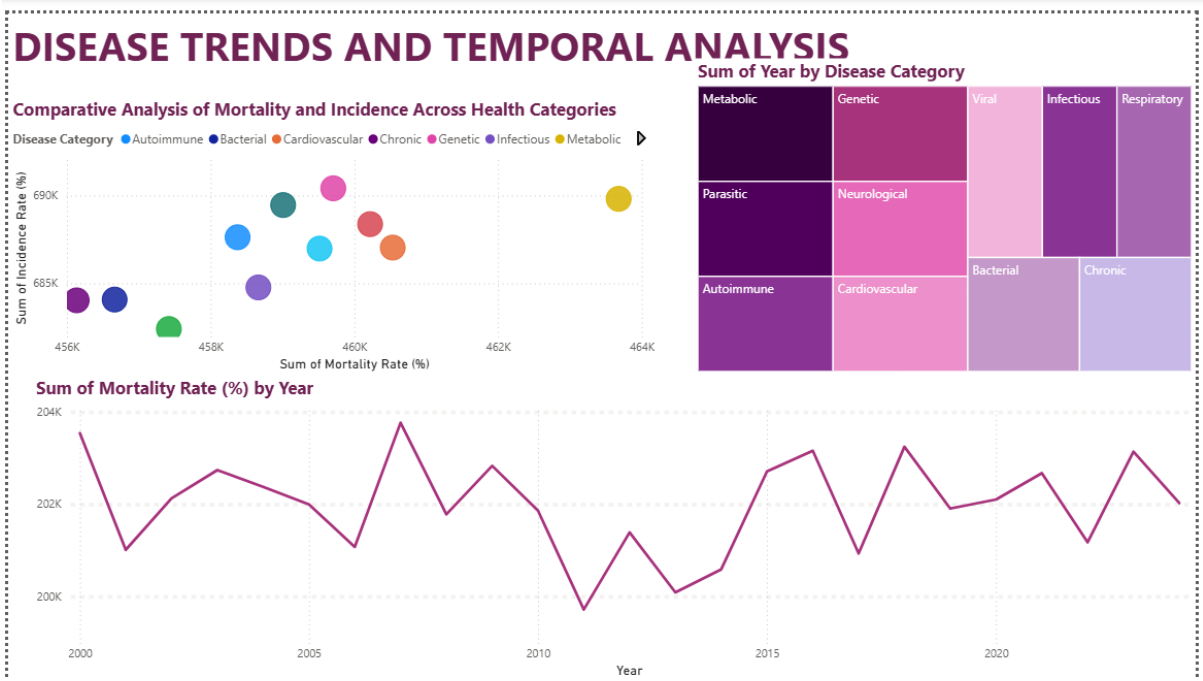
Users can filter the dashboard using:

a. Year

b. Country
c. Disease Name
d. Age Group

These interactive features allow targeted analysis and customized exploration of the dataset.

Page 2



(i) Disease Trends and Temporal Analysis

This focuses on analysing disease behaviour over time and examining relationships between mortality and incidence rates.

(ii) Comparative Analysis of Mortality and Incidence

A scatter plot compares the Sum of Mortality Rate (%) and Sum of Incidence Rate

Key findings include:

- Metabolic diseases appear as outliers with both high mortality and high incidence.
- Genetic and Parasitic diseases also exhibit relatively high incidence rates.
- Chronic and Autoimmune diseases cluster at lower mortality and incidence levels.

High-risk disease categories were identified, and relationships between health indicators understood.

(iii) Disease Category Distribution (Treemap)

The treemap visualizes the proportional distribution of disease categories within the longitudinal dataset.

Large categories include metabolic, genetic, viral and infectious, while smaller categories include chronic and bacterial.

(iv) Mortality Rate Trends Over Time

The line chart displays changes in mortality rates between 2000 and 2024.

The visualization reveals significant fluctuations such as major declines in mortality rates and recovery rates

This temporal analysis supports early detection of emerging health risks and trend monitoring.

Decision support section

a. Resource Allocation

Using the dashboard, health authorities will be able to locate:

i) Areas with high mortality rates
ii) Diseases with rising prevalence
iii) High burden of DALYs (Disability Adjusted Life Year)

This will enhance resource allocation for medical intervention and health care.

b. Early Warning System

Through trend analysis, diseases with fast-rising prevalence or mortality rates can be detected early.

This helps the government take pre-emptive action without the situation getting out of hand in case of an outbreak.

c. Policymaking

The dashboard enhances long-term assessment of policies through comparisons of disease burden between different periods and countries.

Decision-makers can assess intervention effectiveness and design evidence-based health policies.

Human-Computer Interaction (HCI) Considerations

The dashboard was designed using user-centred design principles to improve usability, accessibility, and interpretation.

1. **Simple Layout -** A clean and organized structure reduces cognitive overload and allows users to focus on important insights.
2. **KPI Cards -** KPI cards provide instant summaries of key health indicators, supporting quick decision-making.
3. **Interactive Charts -** Interactive visuals improve exploration and enable users to analyse data dynamically.
4. **Consistent Colours -** Consistent colour usage improves recognition and comparison across charts.
5. **Filters and Slicers -** Filters provide flexibility and allow users to perform focused analysis**.**
6. **Tooltips -** Tooltips provide additional information without overcrowding visuals.
7. **Clear Labels -** Clear titles, legends, and axis labels improve readability and accessibility

Conclusion

The Public Health Dashboard successfully transforms complex health data into interactive and meaningful insights. Through the use of KPI cards, trend analysis, maps, and interactive

visualizations, the system supports healthcare decision-making, policy evaluation, and disease monitoring.